

Amulya Varshney

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LeetCode: **Knight(Rating:1900+)** Codeforces: **Candidate Master (Rating:1900+)** HackerRank: **6 Star**

EDUCATION

- International Institute of Information Technology (IIIT) - Naya Raipur, India** 2019 - 2023
- Bachelor of Technology (B.Tech) - Computer Science and Engineering (CSE)* CGPA: 9.23/10.00

SKILLS

- Languages:** Python, Go, Java, TypeScript, JavaScript, C++, C#, SQL
- AI & LLM:** Agentic AI, RAG, MCP, A2A, A2UI, Agent Memory, Tool Calling, LLM Evaluation, Guardrails, LLMOps
- Framework:** FastAPI, REST, gRPC, GraphQL, Temporal, Kafka, React, Node.js, Next.js, Langchain, LangGraph, Langfuse
- Data & Cloud:** PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Vector DB, Docker, Kubernetes, AWS, Azure, GCP, Linux
- Interest:** Problem Solving, Multi-Agent Systems, Prompt Engineering, Context Engineering, Agent Design, Design Patterns

EXPERIENCE

- AI Software Engineer 2** May 2025 – Present
Bangalore, India
 - Intuit*
 - Designed and built a production-grade **multi-agent AI system** for QuickBooks Payroll & Tax Platform using **LangGraph**, orchestrating planning, retrieval, reasoning, and tool execution to automate complex customer workflows.
 - Architected a scalable **agent orchestration framework** with dynamic routing, tool calling, guardrails, retries, and **RAG** using vector search and domain knowledge, delivering accurate, context-aware responses while reducing hallucinations.
 - Developed persistent **agent memory** and AI infrastructure for conversation awareness, contextual retrieval, and cross-session personalization, enabling seamless multi-turn user interactions.
 - Built **LLMOps** capabilities including **Langfuse**-based observability, prompt tracing, tool analytics, and automated **evaluation pipelines** to measure retrieval quality, routing accuracy, and response correctness before production.
 - Improved **AI platform** performance and reliability through inference optimization, intelligent caching, database tuning, and zero-downtime migrations, ensuring low-latency, highly available payroll-critical AI services.
- Software Development Engineer - AI Platform** Aug 2024 – May 2025
Bangalore, India
 - Razorpay*
 - Built **Ray**, an enterprise AI support agent using **LLMs, LangGraph, FastAPI, and multi-agent orchestration**, automating over **70%** of merchant support requests and reducing response time from hours to seconds.
 - Engineered **multi-agent workflows** with planning, retrieval, execution, and refinement agents, leveraging context-aware prompting and dynamic orchestration to handle complex fintech support scenarios.
 - Developed an internal **LLM evaluation framework in Go** for prompt regression testing, automated quality scoring, safety validation, and continuous benchmarking across model versions.
 - Implemented automated response quality metrics for factual accuracy, instruction adherence, completeness, and financial compliance, improving **AI response quality by 40%**.
- Software Engineer** Jan 2023 – Jul 2024
Bangalore, India
 - Zeiss*
 - Developed secure, scalable backend services using **Java, Spring Boot, REST APIs, and role-based access control (RBAC)**, strengthening authentication and authorization for enterprise security applications.
 - Engineered a real-time **gRPC-based** notification service for virus threat alerts, system updates, and configuration changes, improving incident response time by 60% across distributed systems.
 - Enhanced enterprise applications using **C# and React with TypeScript**, building modular, reusable components and scalable configuration management systems supporting 1,000+ concurrent users.

PROJECTS

- VitalPlan - AI Diet Guide & Nutrition Tracker:** (FastAPI, React, Redis, PostgreSQL, LLMs, Vite, Stripe, Docker, GitHub Actions)
 - Built an **AI-powered nutrition platform** that generates personalized diet plans and food insights using LLMs, integrating multi-modal inputs (text and images) with barcode-based nutrition retrieval for context-aware recommendations.
 - Designed a **provider-agnostic LLM layer** supporting Azure OpenAI, OpenAI, Anthropic, Gemini, and Groq, enabling seamless model switching, structured outputs, and deterministic fallbacks across inference providers.
 - Developed a production-ready **FastAPI + React** application with JWT authentication, Redis caching/rate limiting, CI/CD, Prometheus monitoring, Dockerized deployment, and Stripe integration, following scalable & secure engineering practices.
- Care Gap Closer - Agentic AI for Preventive Healthcare:** (Google ADK, Gemini, LiteLLM, FastMCP, MCP, A2A, FHIR R4)
 - Built an **agentic healthcare platform** that analyzes **FHIR R4** patient records, detects preventive-care gaps using clinical guidelines, and generates personalized patient outreach with LLMs.
 - Designed a **hybrid AI architecture** combining deterministic rule-based reasoning with LLM-powered clinical explanations, and implemented an **MCP server** exposing healthcare tools for secure, structured tool calling.
 - Developed an **A2A agent** using **Google ADK and Gemini** to orchestrate multi-step workflows over **MCP**, with **LiteLLM-based multi-model** support and Dockerized deployment for production-ready scalability.

HONORS AND ACHIEVEMENTS

- Secured **Global Rank 209 and India Rank 37** in Google Kickstart 2022 Round E among 20,000+ Participants.
- Secured **Global Rank 73 and India Rank under 10** in Codeforces contest among 15,000+ Participants.
- Achieved **First Rank** in the CSE department during the first year of graduation with the highest GPA of 9.64.
- Runner Up in Web Dev Hackathon** at HBTU Kanpur Technical Fest out of 10,000+ participants in 2018.
- Fellowship of Science and Technology Council, Uttar Pradesh** shortlisted out of 50,000+ participants in 2017.